BUAN 6356 — Business Analytics
Assignment 4: Pinnacle Outdoor Co.

Total Points: 100
Due Date: 04/15/2025 11:59pm
Deliverable: One R Jupyter notebook (`assignment_4_FirstName_LastName.ipynb`) containing all code, output, and written analysis. Use the **Markdown cells** for the written analysis, interpretations, and recommendations.

Grading emphasis: This is a business analytics assignment, not a machine learning assignment. Points are weighted toward interpretation, business reasoning, and actionable recommendations — not just producing correct code output. A technically correct model with no business interpretation will not receive full marks.

The Company

Pinnacle Outdoor Co. is a mid-size Canadian outdoor recreation retailer founded in 2005. With 30 store locations across Ontario, Alberta, and British Columbia — plus a growing e-commerce channel — Pinnacle sells camping gear, hiking equipment, outdoor apparel, and accessories. Annual revenue is approximately $45 million.

Pinnacle's leadership team faces three pressing business challenges this quarter:

1. **Seasonal inventory clearance.** Summer-season products (tents, hiking gear, outdoor apparel) must sell within a 12-week window. Any unsold inventory at the end of week 12 is liquidated to a discount wholesaler at **25% of the original price**. The CEO wants a data-driven markdown strategy to minimize losses.
2. **Loyalty program churn.** Pinnacle's "TrailRewards" loyalty program has approximately 2,000 active members, but the program is losing members. The VP of Marketing has a limited budget to send **personalized retention offers** (valued at $50 per customer) to at most **20% of members**. A retained customer is estimated to be worth **$1,200 in lifetime value**. She needs to know: *who should receive the offer?*
3. **Ineffective mass marketing.** Pinnacle's email campaigns — sent identically to all customers — achieve less than a 2% response rate. The CMO believes the customer base is diverse and that a one-size-fits-all approach is failing. She wants to understand *who Pinnacle's customers are* so that marketing can be tailored to each group.

The Data

You are provided with two datasets:

`pinnacle_products.csv`

Weekly sales data for 300 seasonal products (SKUs) over a 12-week summer season.

Variable	Description
sku_id	Unique product identifier
product_name	Descriptive product name
category	Product category: Camping, Hiking, Apparel, or Accessories
brand	Brand name
initial_price	Launch price at week 1 (CAD)
current_price	Price in that week (may reflect markdowns)
week	Week number (1–12)
units_sold	Units sold that week
initial_stock	Total starting inventory
cumulative_sold	Cumulative units sold through that week
web_visits	Product page views that week
competitor_price	Competitor's price for a comparable product

pinnacle_customers.csv

Loyalty program member data, including demographics, purchasing behavior, and churn status.

Variable	Description
customer_id	Unique customer identifier
age	Customer age
income	Annual household income (CAD)
household_size	Number of people in household
years_member	Years in TrailRewards program
membership_tier	Bronze, Silver, or Gold
distance_to_store	Distance to nearest store (km)
total_spent_12mo	Total spending in last 12 months (CAD)
num_purchases_12mo	Number of transactions in last 12 months
avg_order_value	Average transaction amount (CAD)
spend_camping	12-month spending on camping products
spend_hiking	12-month spending on hiking products
spend_apparel	12-month spending on apparel
spend_accessories	12-month spending on accessories
pct_online	Proportion of purchases made online (0–1)
days_since_last_purchase	Days since most recent transaction
support_calls	Number of customer service contacts
satisfaction_score	Most recent survey score (1–10)
referred_friend	1 if customer has referred someone, 0 otherwise
churned	1 if customer did not renew membership, 0 otherwise

Both datasets are complete — there are no missing values.

Part 1: Seasonal Pricing Analytics (25 points)

The CEO wants to know which products need markdowns and how aggressive they should be.

1a. (5 points) Compute the **sell-through rate (STR)** at week 6 for all 300 products, where:

$$\text{STR} = \frac{\text{Cumulative Units Sold by Week 6}}{\text{Initial Stock}}$$

Identify the **10 products with the lowest STR**. What do these products have in common? Comment on their category, price range, and any other patterns you observe.

1b. (5 points) For all products with STR below 40% at week 6, compare their current price to the **mean price of products in the same category that achieved STR above 70%**. What does this price comparison suggest about why these products are underperforming?

1c. (8 points) Build a **linear regression** model predicting `units_sold` using `current_price`, `week`, `category`, `web_visits`, and `competitor_price` as predictors. Report the model summary and interpret the coefficients on `current_price` and `web_visits`. Specifically:

- What does the coefficient on `current_price` tell the pricing team about the relationship between price and sales?
- What does the coefficient on `web_visits` suggest about the role of online traffic?

1d. (7 points) Based on your analyses in 1a–1c, recommend a **markdown strategy** for the 10 worst-performing products. Be specific:

- How much should each product's price be reduced (in dollars or percentage)?
- What is the expected revenue if you markdown vs. if you let them liquidate at 25% of initial price?
- Are there any products where liquidation might actually be the better option? Why?

Part 2: Loyalty Churn — Lift Analytics (30 points)

The VP of Marketing has budget to contact 20% of at-risk members with a \$50 retention offer. Who should receive it?

2a. (6 points) Fit a **logistic regression** model predicting `churned` using all available customer features (exclude `customer_id`). Report the model summary. Then:

- Compute the **accuracy** of the model using a 0.5 probability threshold.
- Is this accuracy impressive? Why or why not? (*Hint: consider the base churn rate.*)

2b. (8 points) Score all customers with their predicted churn probabilities. Divide customers into **deciles** (10 equal groups) based on predicted probability, from highest risk (decile 1) to lowest risk (decile 10). For each decile, compute:

- The **actual churn rate**
- The **number of actual churners** captured
- The **lift** over the baseline churn rate

Create a **lift chart** (bar chart of lift by decile) and a **cumulative gains chart** (line chart showing cumulative % of churners captured).

2c. (6 points) Using your decile analysis from 2b:

- If Pinnacle contacts the **top 20%** (deciles 1–2) of predicted churners, what **percentage of all actual churners** does this reach?
- How does this compare to contacting a **random 20%** of members?
- In plain language, what is the business value of using the model vs. random selection?

2d. (10 points) The VP of Marketing needs a **cost-benefit analysis** for the retention campaign. Using the following assumptions:

- Retention offer cost: **$50 per customer**
- Customer lifetime value of a saved customer: **$1,200**
- Probability of saving a contacted churner: **30%**
- Budget: contact **20% of members** (top 2 deciles)

Compute:

1. The **total cost** of the campaign (offers sent to top 20%)
2. The **expected number of churners** in the top 20% (from your decile analysis)
3. The **expected number of saves** (churners contacted × 30% save rate)
4. The **expected revenue from saves** (saves × $1,200 CLV)
5. The **net expected value** (revenue minus cost)

Repeat the same calculation for a **random 20%** selection. Compare the two approaches. Should the VP proceed with the model-targeted campaign? Justify your answer.

Part 3: Customer Segmentation (25 points)

The CMO wants to understand who Pinnacle's customers are and tailor marketing to each group.

3a. (7 points) Select the relevant **numeric features** for clustering (exclude `customer_id`, `churned`, and any non-numeric columns). **Scale** the features and run **Principal Component Analysis (PCA)**.

- How many principal components are needed to capture at least **70% of the variance**?
- Examine the loadings of the **top 2 PCs**. What do they represent? (e.g., "PC1 captures spending level while PC2 captures…")

3b. (6 points) Using the PCA-transformed data, run **K-Means clustering** with $k = 3, 4$, and 5 (use `nstart = 25` for stability). For each k:

- Report the **total within-cluster sum of squares**
- Compute the **average silhouette score**

Plot the **elbow chart** and **silhouette scores**. Which k do you choose and why?

3c. (6 points) Using your chosen k, **profile each cluster**. For each segment, report:

- Mean age, income, and household size
- Mean total spending and top product category
- Mean online purchase percentage
- Give each segment a **descriptive business name** (e.g., "Budget Browsers," "Weekend Warriors")

3d. (6 points) For each segment, recommend a **specific marketing strategy**. For each, specify:

- **Channel**: email, in-store signage, mobile app push, or social media
- **Product category to feature**: which category should be front-and-center?
- **Offer type**: percentage discount, bonus loyalty points, early access to new products, or free shipping

Justify each recommendation using the segment profile from 3c.

Part 4: Executive Synthesis (20 points)

4a. (12 points) Write a **one-page executive memo** to Pinnacle's CEO summarizing your three analyses. For each analysis, state:

1. The **business question** you addressed
2. The **key finding**
3. The **recommended action**

Write for a non-technical audience. No jargon, no model specifications — just clear business language. Use specific numbers where they strengthen the argument.

4b. (8 points) The CEO asks: *"If we can only implement one of your three recommendations this quarter, which should it be and why?"*

Answer with a **quantitative argument**. Consider:

- Which recommendation has the clearest ROI?
- Which addresses the most urgent problem?
- Which is easiest to implement?